CSC 144V: Autonomous Vehicles
Lecture 2

Intro to Machine Learning and Neural Networks Exercises Solution

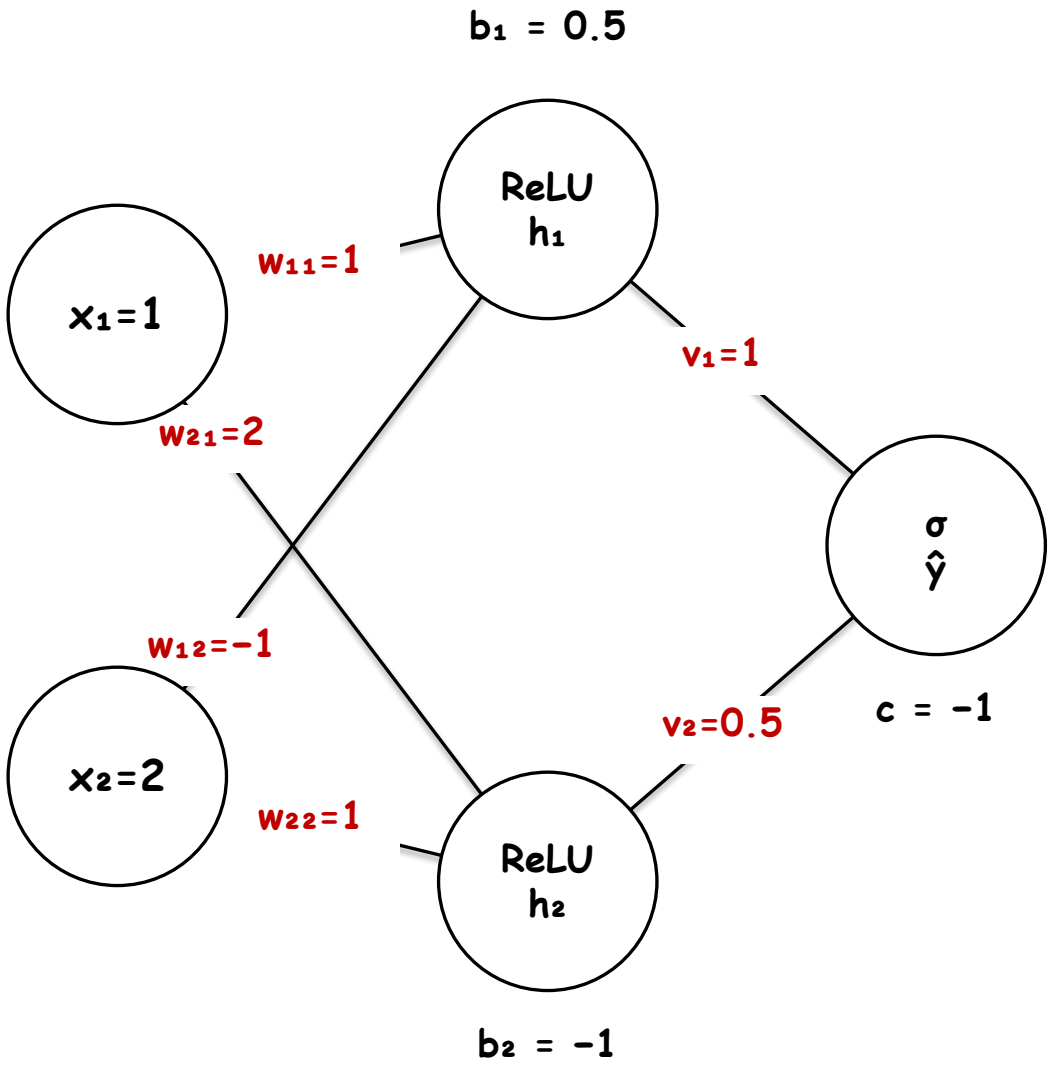
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Exercise: Forward pass

- Input $x = (x_1, x_2) = (1, 2)$, hidden layer: two ReLU neurons, output: sigmoid
  - $z_1 = w_{11}x_1 + w_{12}x_2 + b_1$, $h_1 = \mathrm{ReLU}(z_1)$
  - $z_2 = w_{21}x_1 + w_{22}x_2 + b_2$, $h_2 = \mathrm{ReLU}(z_2)$
  - $z = v_1h_1 + v_2h_2 + c$, $\hat{y} = \sigma(z) = 1/(1 + e^{-z})$
- True label $y = 1$; loss $L = -[y \ln \hat{y} + (1 - y) \ln(1 - \hat{y})]$
- Useful: $\sigma(0.5) \approx 0.622$, $\ln(0.622) \approx -0.474$

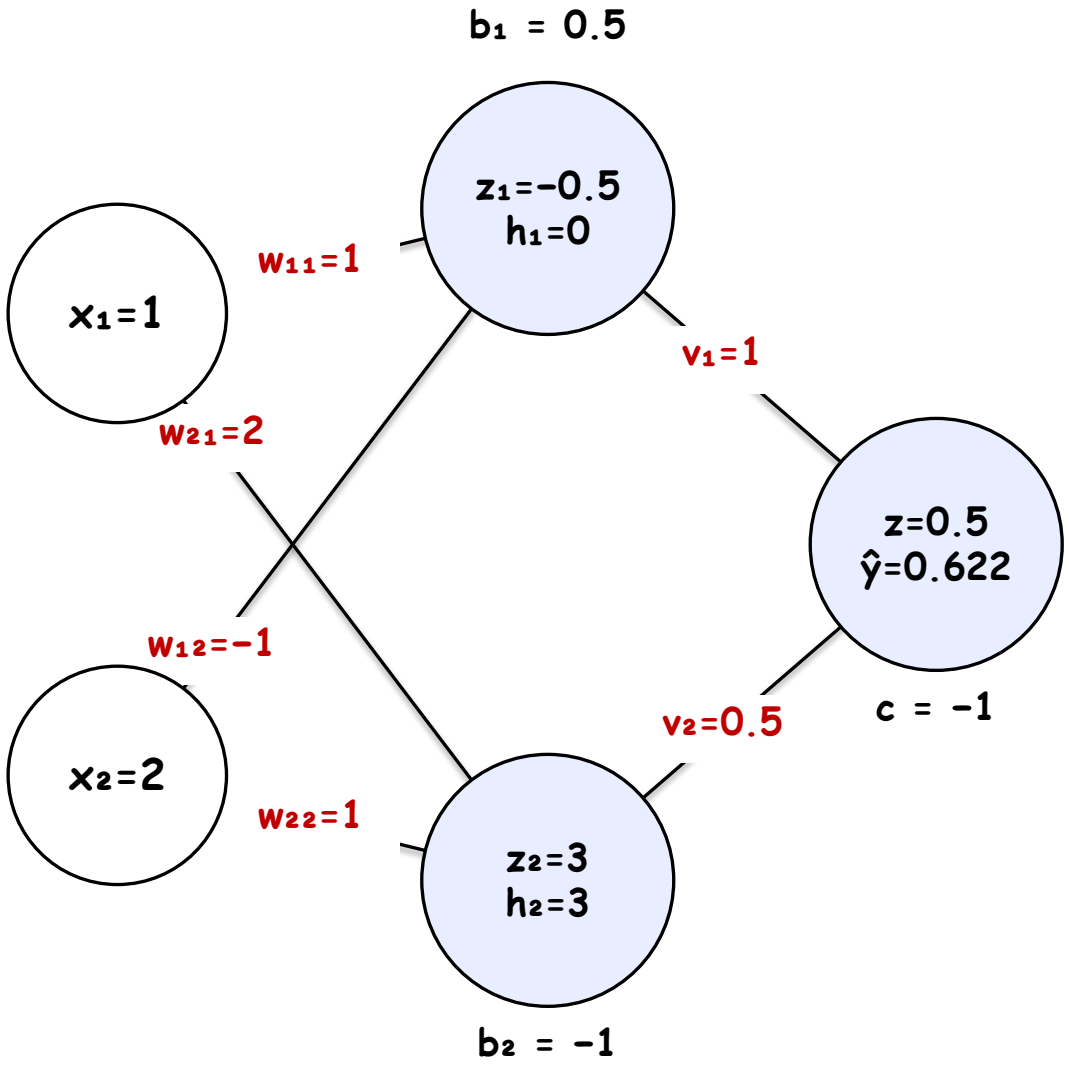
(a) Compute z_1, h_1, z_2, h_2, z and $\hat{y}$.

(b) Compute the loss L.



Exercise Solution: Forward pass

- $z_1 = 1 \cdot 1 + (-1) \cdot 2 + 0.5 = \mathbf{-0.5}$, so $h_1 = \text{ReLU}(-0.5) = \mathbf{0}$
- $z_2 = 2 \cdot 1 + 1 \cdot 2 - 1 = \mathbf{3}$, so $h_2 = \mathbf{3}$
- $z = 1 \cdot 0 + 0.5 \cdot 3 - 1 = \mathbf{0.5}$
- $\hat{y} = \sigma(0.5) \approx \mathbf{0.622}$
- $L = -\ln(0.622) \approx \mathbf{0.474}$

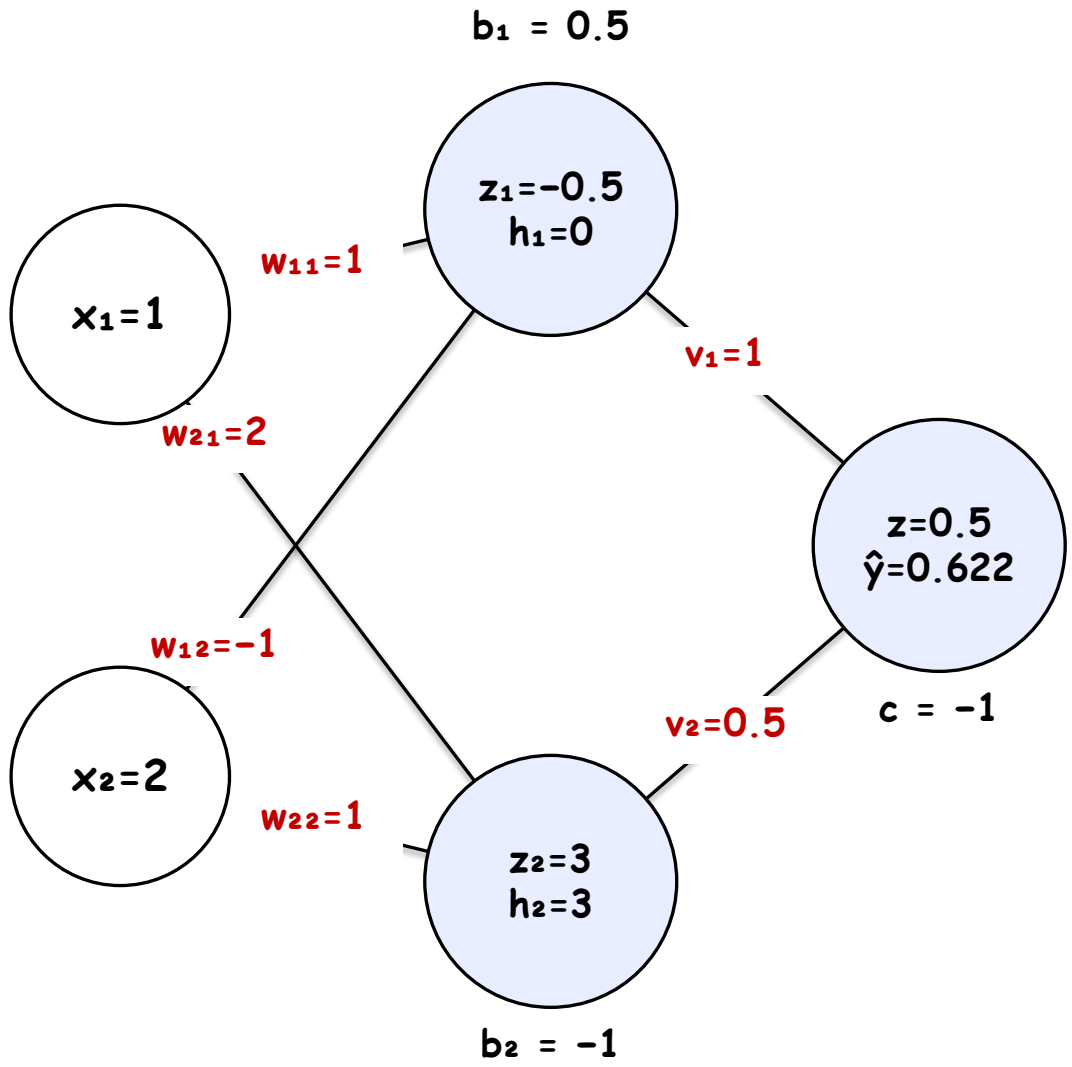


Exercise: Backpropagation

Same network, forward pass as in the previous exercise ($\hat{y} \approx 0.622$, $y = 1$).

(c) For a sigmoid output with binary cross-entropy, $\partial L/\partial z = \hat{y} - y$. Compute $\partial L/\partial z$, $\partial L/\partial v_2$ and $\partial L/\partial c$.

(d) Back-propagate to the hidden layer: compute $\partial L/\partial w_{22}$ and $\partial L/\partial w_{11}$. Explain your result for w_{11} in one sentence.



Exercise Solution: Backpropagation

- $\partial L/\partial z = \hat{y} - y \approx 0.622 - 1 =$ **-0.378**
- $\partial L/\partial v_2 = (\partial L/\partial z)\cdot h_2 = -0.378 \times 3 \approx$ **-1.13**
- $\partial L/\partial c = (\partial L/\partial z)\cdot 1 \approx$ **-0.378**
- $\partial L/\partial h_2 = (\partial L/\partial z)\cdot v_2 = -0.378 \times 0.5 \approx -0.189$; $z_2 = 3 > 0$ so ReLU′ = 1 and $\partial L/\partial z_2 = -0.189$
- $\partial L/\partial w_{22} = \partial L/\partial z_2 \cdot x_2 = -0.189 \times 2 \approx$ **-0.378**
- $\partial L/\partial w_{11} =$ **0**: $z_1 = -0.5 < 0$, so ReLU′(z_1) = 0 — the neuron is inactive for this input and no gradient flows through it (dead ReLU for this sample).

Exercise: Confusion matrix

The same kind of classifier is used as a **pedestrian detector** on 1,000 camera frames, 100 of which contain a pedestrian (positive class).

(e) Compute accuracy, precision and recall.

(f) Compute the F1 score.

(g) If the decision threshold is lowered, how do recall and precision typically change? For an AV, is a false negative or a false positive more dangerous here, and why?

	Predicted +	**Predicted −**
Actual +	TP = 80	FN = 20
Actual −	FP = 45	TN = 855

Metric	**Value**
Accuracy	
Precision	
Recall	
F1	

Exercise Solution: Confusion matrix

- Accuracy = (80 + 855)/1000 = **93.5 %**; Precision = 80/(80 + 45) = **0.64**; Recall = 80/(80 + 20) = **0.80**
- F1 = 2PR/(P + R) = 2 × 0.64 × 0.8/1.44 ≈ **0.711** (= 2TP/(2TP + FP + FN) = 160/225)
- Lowering the threshold: **recall increases** (fewer FN), **precision typically decreases** (more FP).
- A **false negative** (missed pedestrian) is more dangerous: the AV does not brake. A false positive usually causes only unnecessary braking, so safety-critical detectors favor high recall.
- 93.5 % accuracy looks good although 36 % of the detections are false alarms.

	Predicted +	**Predicted −**
Actual +	TP = 80	FN = 20
Actual −	FP = 45	TN = 855

Metric	**Value**
Accuracy	**93.5 %**
Precision	**0.64**
Recall	**0.80**
F1	**0.711**